

P.1: The Districting Integrity Act: Legislative History, Constitutional Authority, and Alternatives

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Abstract

Congress possesses authority under Article I, §4 to regulate the “Manner” of congressional elections, including the drawing of district lines. We analyze the Districting Integrity Act (DIA), a model federal statute requiring states to use recursive bisection to produce their congressional maps, and compare it against four alternative federal legislative designs: algorithm-specifying statutes, criteria-only statutes, commission-mandating statutes, and court-remedy statutes. The DIA’s architecture is deliberately analogous to the Apportionment Act of 1941 (now codified at 2 U.S.C. §2a), which mandated the Huntington-Hill method for apportioning seats among states. Like that statute, the DIA prescribes a specific deterministic algorithm—recursive bisection as implemented in the `bisect` platform—produces verifiable outputs, and preserves state responsibility for execution while preempting partisan manipulation. We assess each alternative design against four criteria: constitutional durability, political feasibility, VRA compatibility, and technical implementability. The DIA ranks highest overall. The criteria-only design (ranking second) provides a useful fallback if Congress lacks the technical expertise or political will to specify an algorithm. The commission-mandating and court-remedy designs are complements to, not substitutes for, an algorithm-based approach.

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1 Introduction

In *Rucho v. Common Cause*, 588 U.S. 684 (2019), the Supreme Court held that partisan gerrymandering claims are nonjusticiable in federal court because “federal courts are not meant to resolve such political questions.” But the Court also emphasized what it did not hold: it did not hold that partisan gerrymandering is acceptable, or that no legal remedy exists. “[N]othing in this opinion should be interpreted as expressing support for partisan gerrymandering,” Chief Justice Roberts wrote. “Of course,” the opinion continued, “the limits of the federal courts’ authority does not leave Americans without a remedy for extreme partisan gerrymandering. The States may regulate congressional elections; they may also provide independent state courts with authority to review legislatively drawn plans for compliance with state constitutional requirements.” Most pointedly: “[F]ederal courts are not the only potential check on partisan gerrymandering. Indeed, the plurality noted the ‘daunting’ challenges federal courts have faced in attempting to adjudicate partisan gerrymandering claims. But the limitations of the judicial role in no way suggest that partisan gerrymandering is permissible.” [\[Supreme Court of the United States, 2019\]](#)

The implication is direct: Congress may act. Article I, §4 of the Constitution grants Congress authority to regulate the “Times, Places and Manner” of congressional elections, and Congress may override or supplement state regulations. Congress has previously exercised this power to require single-member districts (2 U.S.C. §2c), to prohibit certain election-day practices, and to require equal population among congressional districts in application of *Wesberry v. Sanders*, 376 U.S. 1 (1964).

The model federal statute analyzed in this paper—the Districting Integrity Act (DIA)—takes its structural cue from the most successful prior exercise of Congress’s redistricting authority: the Apportionment Act of 1941, which mandated the Huntington-Hill equal-proportions method for apportioning House seats among states after each census. That statute resolved a decades-long dispute over apportionment by specifying a single deterministic algorithm whose output any person can verify. The DIA applies the same logic to the drawing of district lines within states.

Section 2 analyzes the constitutional authority for federal redistricting legislation. Section 3 describes the DIA’s key architectural choices. Section 4 compares the DIA against four alternative designs. Section 5 develops the historical analogy to the Apportionment Act of 1941 in depth. Section 6 concludes with implications for federal legislative strategy.

The full text of the DIA model statute is maintained in `docs/legal/MODEL_FEDERAL_STATUTE.md` as part of the `bisect` research program’s documentation.

2 Constitutional Authority

The constitutional authority for federal redistricting legislation is well established, though its precise scope has been contested.

2.1 Article I, §4: The Elections Clause

Article I, §4, cl. 1 provides: “The Times, Places and Manner of holding Elections for Senators and Representatives, shall be prescribed in each State by the Legislature thereof; but the Congress may at any time by Law make or alter such Regulations, except as to the Place of chusing Senators.” This clause is the primary constitutional basis for the DIA.

The Elections Clause’s breadth is not in doubt. In *Ex parte Siebold*, 100 U.S. 371 (1880), the Supreme Court held that Congress may supplement, displace, or correct state election regulations. In *Smiley v. Holm*, 285 U.S. 355 (1932), the Court confirmed that Congress could constitutionally mandate that states draw congressional districts as single-member geographic units. In *Cook v. Gralike*, 531 U.S. 510 (2001), the Court clarified that the clause grants states authority to regulate “the procedural mechanisms for obtaining a ballot position,” and grants Congress equivalent authority to override state rules.

The Elections Clause question most relevant to the DIA is whether Congress may specify a particular method—not merely criteria—for how states draw their lines. The apportionment analogy (Section 5) suggests yes: 2 U.S.C. §2a specifies the Huntington-Hill algorithm with mathematical precision, and it has never been constitutionally challenged. The DIA applies the same structural template to within-state districting.

2.2 Section 5 of the Fourteenth Amendment

The DIA also asserts authority under Section 5 of the Fourteenth Amendment, which grants Congress power to enforce the Amendment’s requirements by appropriate legislation. *Reynolds v. Sims*, 377 U.S. 533 (1964), established that the Equal Protection Clause requires substantially equal population among congressional districts. Congress may enforce this requirement through legislation.

The dual-authority design in DIA §101(f) is a defense against the risk that a future Court narrows the Elections Clause. If either authority is sufficient—Article I or Fourteenth Amendment §5—the statute survives even if one basis is rejected. This design follows the pattern of the Voting Rights Act, which also asserts dual authority under the Fifteenth Amendment and the Fourteenth.

2.3 The Printz/Murphy Anti-Commandeering Limit

The most significant constitutional risk for the DIA is the anti-commandeering doctrine. *Printz v. United States*, 521 U.S. 898 (1997), holds that Congress may not “commandeer” state officers to administer federal programs. *Murphy v. NCAA*, 138 S. Ct. 1461 (2018), extended this principle to prohibit Congress from “dictating” that states maintain a particular regulatory regime.

The DIA’s rebuttal is textual: the Elections Clause *expressly* authorizes federal alteration of state election regulations. As DIA §101(g) explains, the Tenth Amendment does not constrain powers the Constitution affirmatively assigns to Congress. This argument is supported by *Foster v. Love*, 522 U.S. 67 (1997), which upheld a federal statute overriding Louisiana’s open-primary system, and by *Branch v. Smith*, 538 U.S. 254 (2003), which allowed federal courts to draw congressional maps when states failed to timely comply with federal requirements.

The anti-commandeering risk can be further reduced by structuring the DIA as a direct federal mandate (addressed to states as sovereign entities acting in their governmental capacity) rather than as a directive to individual state officers. This distinction matters under *Printz*’s framework.

2.4 Non-Delegation Concerns

The DIA delegates to the Director of the Census Bureau the authority to specify certain technical parameters—the seed for the algorithm, the version of the reference binary, and the TIGER adjacency errata process. This delegation is narrowly circumscribed: the statute itself specifies the algorithm (recursive bisection), the population metric (decennial census counts), the balance tolerance ($\pm 0.5\%$), and the edge-weight specification. The Director’s authority extends only to release-specific administrative decisions that cannot be fixed in statute.

Under *West Virginia v. EPA*, 597 U.S. 697 (2022)’s major questions doctrine, regulations of significant political importance require clear congressional authorization. The DIA provides that authorization explicitly, by name, with specific parameter ranges. The delegation is ministerial, not policymaking—analogous to the Census Director’s existing authority to specify the format of decennial data releases.

3 The DIA’s Architectural Design

The DIA is structured around five core architectural choices, each of which responds to a specific legal or political constraint.

3.1 Algorithm Specification, Not Criteria

The DIA specifies a particular algorithm—recursive bisection with METIS edge-cut optimization—rather than criteria that a map must satisfy. This is the DIA’s most distinctive and controversial design choice. Alternative designs that specify only criteria (contiguity, compactness, population equality) leave substantial room for partisan manipulation within those criteria: a skilled mapmaker can produce maps that satisfy all listed criteria while still dramatically favoring one party.

The algorithm-specification approach eliminates this residual manipulation space. Once the algorithm, its inputs, and its parameters are fixed, the output is deterministic. There is no manipulation space because there is no human mapmaker making choices about how to satisfy the criteria.

The precedent for algorithm specification is the Apportionment Act of 1941, discussed in Section 5. Congress specified not merely that apportionment should be “proportional” but that it should use the equal-proportions method with specific mathematical formulation. This level of specificity is both constitutionally permissible and practically necessary to prevent gaming.

3.2 Byte-Reproducibility as the Verification Standard

The DIA’s verification standard is byte-reproducibility: any person, using the reference binary distributed by the Census Bureau, can run the algorithm on public inputs and produce a map that is identical, byte-for-byte, to the officially adopted plan. This standard is more demanding than the “reasonable person” standard common in administrative law, but it is achievable and it resolves disputes mechanically rather than judicially.

The manifest format in the `bisect` platform implements this standard. The manifest records the binary version (SHA-256 hash), the input adjacency file (SHA-256 hash), and the census release identifier. Any party can verify a plan by:

```
bisect label-verify <label> --year 2020
```

If verification fails, the plan is presumptively invalid. If it succeeds, the plan satisfies the statutory requirement. Courts do not need to evaluate cartographic expertise; they evaluate a hash.

3.3 VRA Preservation: The Modification Protocol

The DIA requires that the algorithm’s output serve as the baseline from which VRA-required modifications are made. Where *Thornburg v. Gingles*, 478 U.S. 30 (1986), and the three-part test it established compel creation or preservation of majority-minority districts, the state may—and under DIA §106(b), *shall*—modify the baseline to comply.

Critically, each modification must be documented: the specific *Gingles* prongs satisfied, the minority community affected, and the tract-level changes made. This documentation requirement serves two purposes. It limits modification to VRA-required adjustments (preventing use of the VRA exception as a general escape hatch) and it creates a record for subsequent litigation. Under *Allen v. Milligan*, 599 U.S. 1 (2023), Section 2 of the VRA remains a robust constraint; the DIA’s modification protocol implements that constraint in a verifiable way.

3.4 State Execution, Federal Floor

The DIA does not draw maps for states. States retain responsibility for running the algorithm, publishing the outputs, and managing the VRA modification process. The federal government distributes the reference binary, maintains the parameter table, and provides a safe harbor against liability for binary defects. This structure preserves state sovereignty in execution while establishing a federal floor below which states cannot fall.

The “safe harbor for binary defects” provision (DIA §107(d)) addresses a practical concern: if the reference binary has a bug, states that relied on it in good faith should not be penalized. This provision is necessary for states to trust the federal binary and is analogous to good-faith reliance doctrines in tax and regulatory contexts.

3.5 Timeline and Enforcement

The DIA requires publication of the baseline plan within 180 days of receiving the decennial census redistricting data for states with eight or fewer congressional districts, and within 270 days for larger states. This tiered timeline reflects the difference in computational and administrative complexity between small and large states.

Enforcement is through private right of action in three-judge district courts, with asymmetric fee-shifting: plaintiffs who successfully enforce VRA-protective provisions recover fees; plaintiffs who challenge the algorithm itself on partisan grounds do not. A 30-day cure period applies to non-VRA technical defects, giving states opportunity to correct errors without immediate litigation exposure. Criminal penalties apply to any state official who attempts to

substitute a politically drawn map for the algorithm’s output.

4 Four Alternative Designs

We evaluate the DIA against four alternative federal legislative designs. Each has genuine merits and specific weaknesses.

Table 1: Alternative federal redistricting bill designs compared

Design	Constitutional Durability	Political Feasibility	VRA Compat.	Technical Implement.
DIA (algorithm-specifying)	High	Medium	High	High
Criteria-only	High	High	Medium	Low
Commission-mandating	Medium	Low	High	Medium
Court-remedy	High	High	High	Medium

4.1 Design A: Algorithm-Specifying (the DIA)

The DIA specifies recursive bisection as the redistricting algorithm. As analyzed in Section 3, this design is constitutionally strong, technically implementable, and VRA-compatible with the modification protocol. Its primary limitation is political feasibility: many members of Congress—particularly those from safe seats—are unlikely to vote for legislation that could make their districts more competitive, and the algorithm specification is novel enough to generate concerns from members who are skeptical of “computer-drawn maps.”

The political barrier can be partially mitigated by framing: the DIA does not eliminate human judgment, it structures it. The algorithm produces a neutral baseline; humans apply the VRA modification protocol. This framing is accurate and politically more accessible than “the computer draws the maps.”

4.2 Design B: Criteria-Only

A criteria-only statute specifies the requirements a map must meet—contiguity, population equality, compactness, political subdivision respect—without specifying how the mapmaker achieves those criteria. The For the People Act (H.R. 1 in several Congresses) adopted this general approach, requiring independent commissions and listing criteria but leaving the drawing method to the commission.

The criteria-only design has high political feasibility because it is less technically specific and avoids “computer draws maps” concerns. Its weakness is technical: criteria-only statutes

do not eliminate partisan manipulation. A sufficiently skilled mapmaker can satisfy all listed criteria while still maximizing partisan advantage—this is precisely what North Carolina did in the maps challenged in *Rucho*. The map satisfied all state-law criteria; it was also, the district court found, “one of the largest partisan gerrymanders in modern American history.”

Criteria-only statutes can be strengthened by adding measurable thresholds (e.g., no map with an efficiency gap above 7% may be adopted) or by requiring the criteria-satisfying map to minimize a neutral objective function. But once you have a neutral objective function and a minimization requirement, you are most of the way to an algorithm-specifying statute.

4.3 Design C: Commission-Mandating

A commission-mandating statute requires states to create independent redistricting commissions but does not specify how commissions draw maps. The model here is the For the People Act’s commission provisions, which establish structural requirements (balanced partisan representation, no elected official members, public hearing requirements) without specifying a drawing methodology.

Commission-mandating statutes face a significant constitutional uncertainty: *Rucho* is a federal-court case, but several state courts have held that state constitutional provisions vesting legislative authority in the “legislature” preclude voter-initiative commissions. *Arizona State Legislature v. Arizona Independent Redistricting Commission*, 576 U.S. 787 (2015), upheld the Arizona commission against a federal Elections Clause challenge, but the decision was narrow and a future Court could reconsider it. Congress mandating commissions in all states raises the same structural question at the federal level.

Commission-mandating statutes also face political feasibility challenges: members of Congress who draw their own districts are unlikely to vote for legislation requiring them to cede that authority to commissions, even commissions with balanced partisan representation. At least an algorithm-specifying statute might be framed as being “fair to both parties”; a commission adds a layer of partisan negotiation over the commission’s composition and operating rules.

4.4 Design D: Court-Remedy Statute

A court-remedy statute does not mandate a particular redistricting method but provides federal courts with a clear legal standard and remedy when states fail to meet it. The statute would define actionable partisan gerrymandering (e.g., any map with a two-standard-deviation efficiency gap outlier), establish a judicial remedy (the federal court appoints a special master who uses the `bisect` algorithm), and provide expedited procedures to reduce the delay

problem identified in P.4.

This design is politically feasible because it does not mandate any particular state action in the absence of litigation—states that draw fair maps face no constraint. It is constitutionally strong because it simply creates a federal cause of action, a traditional judicial function. Its weakness is that it preserves the same slow, expensive litigation dynamic that currently characterizes redistricting disputes. It helps when litigation occurs but does not prevent the underlying gerrymandering.

A court-remedy statute could be combined with either the DIA or the criteria-only design: the substantive mandate establishes what states must do, and the court-remedy provisions specify what happens when they don't. P.4 analyzes court-ordered algorithmic remedies in depth.

4.5 Selecting Among Designs

For Congress that lacks the political will for the DIA, we recommend the criteria-only design with quantitative thresholds as a second-best option. Thresholds—efficiency gap bounds, minimum compactness scores—are more gameable than an algorithm but substantially less gameable than qualitative criteria. A court-remedy statute should be included in any design to provide enforcement mechanisms.

The DIA remains the preferred design because it eliminates the residual manipulation space present in all criteria-based approaches. The algorithm-analogy to Huntington-Hill provides the most powerful historical precedent, and the technical implementation—as demonstrated by the `bisect` platform—is fully mature. The political barrier is real but surmountable; Section 6 suggests a political strategy.

5 Historical Precedent: The Apportionment Act of 1941

The most powerful precedent for the DIA is not a redistricting case but an apportionment statute: the Act of November 15, 1941 (55 Stat. 761), now codified at 2 U.S.C. §2a(a), which mandated the Huntington-Hill equal-proportions method for allocating House seats among states after each decennial census.

5.1 The Problem the 1941 Act Solved

Before 1941, Congress had repeatedly failed to apportion House seats after the 1920 census. The dispute was not about population data—the census was uncontested—but about the method. Different mathematical formulas (Webster’s major fractions, Hamilton’s largest remainders, Dean’s harmonic mean) produced different allocations in closely contested cases, and each method happened to favor different regional or partisan coalitions. Congress deadlocked for a decade.

The impasse was resolved by algorithm specification. The National Academy of Sciences was asked to evaluate competing methods on neutral mathematical grounds. It recommended Huntington-Hill equal-proportions as the method that minimized relative representation differences among states. Congress enacted that specific method in 1941 and has not changed it since. The statute is brief—the relevant operative language is fewer than 200 words—but it is mathematically precise.

5.2 Structural Parallels to the DIA

The Huntington-Hill statute and the DIA share five structural features:

(1) Algorithm rather than criteria. The 1941 Act does not say that apportionment should be “proportional” or “fair.” It specifies the exact mathematical operation: assign $a_i = \sqrt{P_i / (n_i \cdot (n_i + 1))}$ for each state i , where P_i is population and n_i is the current number of seats, then assign the next seat to the state with the highest priority value. The DIA likewise specifies recursive bisection with METIS edge-cut optimization, not merely “compact and equal-population districts.”

(2) Delegated technical parameters. The 1941 Act delegates to the Census Bureau the responsibility for applying the formula and certifying the results. The DIA delegates to the Census Director the authority to specify the reference binary version and seed. Both delegations are ministerial, not policymaking.

(3) Verifiable output. Huntington-Hill apportionment is verifiable by any person with a spreadsheet. The DIA’s bisection output is verifiable by any person with the reference binary. In both cases, verification is mechanical, not expert-dependent.

(4) Preserved state role. The 1941 Act determines how many seats each state receives; states then draw their own district lines. The DIA determines how district lines are drawn; states manage the VRA modification process. In both cases, the federal law establishes a floor and delegates execution to states.

(5) Successful congressional resolution of a “political” question. The apportionment dispute was, like partisan gerrymandering, a question where interested parties

had strong stakes in the outcome. Congress resolved it by delegating to a mathematical algorithm. The DIA applies the same political judo: rather than asking Congress to pick between competing partisan visions of fair maps, it asks Congress to pick an algorithm that has no partisan vision at all.

5.3 The 1941 Act’s Constitutional Track Record

The Huntington-Hill statute has never been successfully challenged on constitutional grounds. It has been applied in every apportionment since 1950. It has survived challenges arguing that other methods would produce more proportional outcomes (*United States Department of Commerce v. Montana*, 503 U.S. 442 (1992), which rejected a constitutional mandate for perfect proportionality) and that the method disadvantaged particular states (*Department of Commerce v. U.S. House of Representatives*, 525 U.S. 316 (1999), upholding the method but not the proposed use of statistical sampling).

The statute’s constitutional durability over 80 years, through multiple party changes in Congress and multiple Supreme Court compositions, is the best available evidence that algorithm-specification is within Congress’s Elections Clause authority. The DIA cites this precedent explicitly in its findings section (§101(c)).

5.4 Limitations of the Analogy

The analogy is powerful but imperfect. Apportionment allocates seats among states (a federal problem with states as the affected parties) while redistricting allocates voters among districts within a state (a state problem with individuals as the affected parties). The constitutional bases differ—apportionment is grounded in Article I, §2; redistricting is grounded in Article I, §4 and the Fourteenth Amendment.

The analogy also breaks down in one practical respect: Huntington-Hill’s inputs are minimal (state populations) while bisection’s inputs are extensive (tract adjacency graph, population data). The `bisect` platform manages this complexity automatically via:

```
bisect fetch --year 2020 --workers 8
bisect build <label> --year 2020
```

But the greater input complexity creates more surface area for disputes about data quality, adjacency definitions, and boundary conditions. The DIA addresses this by locking the TIGER adjacency graph and census population data to specific federal releases, with an errata mechanism for corrections.

Despite these limitations, the 1941 analogy provides the strongest available historical argument for the DIA’s constitutional validity and political precedent. Congress has done this before. It can do it again.

6 Conclusion

The Districting Integrity Act represents the most complete available solution to partisan gerrymandering: federal legislation that eliminates partisan manipulation at its source by specifying a neutral algorithm whose output any person can verify. Its constitutional authority is grounded in Article I, §4 and Fourteenth Amendment §5. Its technical implementation is complete, available as the open-source **bisect** platform. Its historical precedent—the Apportionment Act of 1941—demonstrates that Congress has successfully resolved analogous disputes by delegating to mathematics.

The DIA outperforms all four alternative designs on the combined dimensions of constitutional durability, VRA compatibility, and technical implementability. It ranks below the criteria-only and court-remedy designs on political feasibility, but this disadvantage is a function of current political conditions rather than intrinsic design defects. A criteria-only bill with quantitative thresholds is a viable second-best option for Congress that lacks the votes for full algorithm specification.

The primary political barrier is incumbency self-interest. Members of Congress who benefit from safe districts are unlikely to vote for legislation that makes those districts competitive, regardless of the reform’s merits. This barrier, analyzed in P.0, is real. But it is not insurmountable. The reform coalition must be built from members who are in competitive districts and who have less to lose from neutral maps, combined with members who represent geographically concentrated partisan minorities who are currently systematically underrepresented.

The timing window for federal action is 2025–2030. Redistricting for the 2030 census will occur in 2031. Federal legislation enacted by 2028 would give states two to three years to implement the **bisect** platform before the next redistricting cycle. Implementation is not technically complex—a single **bisect build** command produces the baseline plan in under 20 minutes—but states will need time to train personnel, conduct audits, and develop public engagement processes. Early enactment is essential.

If the DIA is enacted, the redistricting fights of 2031 will be a fraction of the battles of 2021. The algorithm will produce the baseline. VRA modifications will be documented and verifiable. Courts will review manifests, not cartographic expertise. The political energy that currently goes into gerrymandering will go somewhere else. That is the DIA’s ultimate

promise: not a perfect map, but a map whose imperfections are grounded in legal requirements rather than partisan strategy.

References

Supreme Court of the United States. *Rucho v. Common Cause*, 588 U.S. 684, 2019.